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Global Bifurcation Analysis of a Controlled Underactuated Mechanical System

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Abstract. In this paper, bifurcation theory is employed to classify different dynamical behaviors arising in an underactuated mechanical system subject to bounded controls. The methodology is applied to an inertia wheel pendulum consisting of a simple pendulum with a rotating disk at the end. Restricting the magnitude of the control action places an important obstacle to the design of a continuous controller capable of swinging-up and stabilize the pendulum at the inverted position: the arm only can reach that position by means of oscillations of increasing amplitude. The controller is derived from a simple nonlinear state-feedback law, followed by a saturating device that limits the maximum amplitude of the control action applied to the system. This bound gives birth to a rich dynamical behavior, including pitchfork and Hopf bifurcations of equilibria, saddle-node bifurcations of periodic orbits, homoclinic and heteroclinic bifurcations. The global dynamics is analyzed in terms of certain control gains and a two-parameter bifurcation diagram is derived. It is shown that the dynamics on this bifurcation diagram is organized in a pair of codimension-two rotationally symmetric bifurcation points. Finally, it is found out that when the control gains lie on a certain region in the parameter space simultaneous stabilization of the upright position together with a large basin of attraction is obtained. Simulation results show that almost global stabilization of the system can be achieved.

Key words: bounded control, bifurcation theory, stabilization, underactuated mechanical systems

1. Introduction

Simple mechanical systems, such as the pendulum, are frequent examples in the design of stabilizing controllers for nonlinear systems. One of these devices is the inertia wheel pendulum consisting in a simple pendulum with a rotating disk at the end (see Figure 1). This underactuated mechanical system has relatively simple dynamics compared to other systems of the same class, but presents many interesting problems arising in the control of unstable nonlinear plants.

An effective, widely used and somewhat intuitive method to solve the stabilization problem is based on a dual controller strategy, using a nonlinear controller to swing-up the pendulum and then switching to a linear controller to locally stabilize the system at the inverted position. This technique has been applied to a wide class of inverted pendula (see, for example, [1–6]). For the inertia wheel pendulum, feedback linearization and passivity-based control are discussed in [7] in order to solve both stages of control. Other techniques such as anticontrol of Hopf bifurcations or energy control can also be applied to perform the swing-up (see [8]).

In contrast to the switched strategy, the stabilization of underactuated mechanical systems with a unique continuous feedback law presents serious obstacles for the controller design [9]. In [10, 11] almost global[1] stabilizing continuous controllers for the inertia wheel pendulum have

[1] Almost global means that the set of initial conditions which do not converge to the upward position has zero measure.

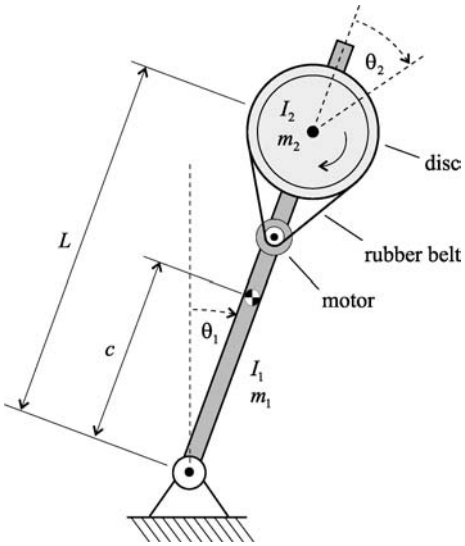


Figure 1. The inertia wheel pendulum.

been derived. It is worth mentioning that for topological reasons global stabilization can not be achieved.

The design of a continuous controller is harder when the amplitude of the control action is bounded. This problem is addressed in [12] for the inertia wheel pendulum, but the proposed controller can not guarantee the stabilization of the inverted position when the available control torque is smaller than the gravity torque. This drawback is also present in the controllers reported in [10, 11].

In this paper such obstacle is removed, and it is shown that a controller with restricted output can achieve stabilization at the upward position when the control torque is smaller than the gravity torque. Using tools from bifurcation theory, the local and global dynamical behavior of the system controlled by a simple continuous and bounded controller (first proposed in [13]) is studied. It is shown that a small control amplitude does not necessary imply a reduced domain of attraction of the equilibrium point. Both analytical and numerical methods are used to classify the different dynamical scenarios arising when control parameters are varied. It is shown that, when considering a two-parameter plane, the system dynamics is organized in a pair of codimension two bifurcations points with rotational symmetry, also known as Khorozov–Takens bifurcation. A partial global unfolding of this bifurcation allows us to identify a region of the considered parameter plane where the stabilization of the pendulum at the inverted position together with a large basin of attraction is achieved. In fact, this analysis encourages us to conjecture that the stabilization is almost global.

Bifurcation theory has been used recently in the context of controlling mechanical systems to describe different (local) dynamical behaviors of a 2-DOF robot manipulator driven by constant torques [14]. Furthermore, local and global bifurcations in a rotational inverted pendulum have been studied in [15, 16], respectively, and the domain of attraction of the stabilized equilibrium has been analyzed in [17] using bifurcation analysis and Lyapunov functions. The problem of controlling a pendulum on a cart in the upright position by applying a delayed horizontal control force is studied in [18]. It is found that a codimension-three (triple-zero eigenvalue) bifurcation is the organizing centre of the dynamics. This bifurcation is also present in the problem considered here and it is the true organizing centre for the dynamics. Nevertheless, we restrict the analysis to the codimension two bifurcation since the triple zero singularity is beyond the scope of this paper. As is the case in all the mentioned papers, the methodology

presented here can be helpful to obtain new insights on the system dynamics to design more efficient controllers for mechanical systems.

2. The Inertia Wheel Pendulum

The equations of motion of the inertia wheel pendulum are easily obtained using the Euler–Lagrange formalism considering as generalized coordinates the angular positions of the pendulum (θ_1) and the disk (θ_2), referred to the vertical line and the arm, respectively. The Lagrangian is

$$L = \frac{1}{2} I_{1\text{eq}} \dot{\theta}_1^2 + \frac{1}{2} I_2 (\dot{\theta}_1 + \dot{\theta}_2)^2 - Mg(1 + \cos\theta_1),$$

with $I_{1\text{eq}} := m_1 c^2 + m_2 L^2 + I_1$, $M := (m_1 c + m_2 L)$, where m_1, I_1 and m_2, I_2 are the mass and the moment of inertia of the arm and disk, respectively; L and c are the distances from the pivot point to the disk shaft and to the center of mass of the arm, respectively; and g is the acceleration of gravity (see Figure 1).

A simplified but accurate model of the torque applied by the DC-motor to the disk is

$$\tau = \frac{N k_T}{R} V_i - \frac{N^2 k_T k_F}{R} \dot{\theta}_2,$$

where N is the motor to disk reduction ratio (due to the coupling by a rubber belt), R is the rotor coil resistance, k_T is the torque constant, k_F is the back-emf constant, and V_i is the control voltage. The electric time constant of the motor has been neglected since it is generally much smaller than the mechanical time constant.

The equations of motion are given by

$$\begin{aligned} (I_{1\text{eq}} + I_2)\ddot{\theta}_1 + I_2 \ddot{\theta}_2 - Mg \sin\theta_1 &= 0, \\ I_2 \ddot{\theta}_1 + I_2 \ddot{\theta}_2 &= \tau. \end{aligned}$$

As the equations of motion do not depend on the disk position (it is a cyclic coordinate [19]), a reduced model of the system can be obtained defining the states variables $x_1 = \theta_1$, $x_2 = \dot{\theta}_1$ and $x_3 = \dot{\theta}_2$, and the control input $u = V_i/\beta$, where β is a positive parameter that accounts for the gain of the motor driver. The three-dimensional model is then

$$\begin{aligned} \dot{x}_1 &= x_2, \\ \dot{x}_2 &= p_1 \sin x_1 + p_2 x_3 - p_3 u, \\ \dot{x}_3 &= -p_1 \sin x_1 - p_2(1 + \rho) x_3 + p_3(1 + \rho) u, \end{aligned} \tag{1}$$

where

$$p_1 := \frac{Mg}{I_{1\text{eq}}}, \qquad p_2 := \frac{N^2 k_T k_F}{I_{1\text{eq}} R}, \qquad p_3 := \frac{\beta N k_T}{I_{1\text{eq}} R}, \qquad \rho := \frac{I_{1\text{eq}}}{I_2},$$

are positive parameters.

3. Design of the Control Law and Local Bifurcation Analysis

The equilibrium points of the unforced system (1) are $x_e = (x_{1e}, 0, 0)$, where x_{1e} is $x_{10} = 2k\pi$ (inverted or upward position) or $x_{1\pi} = (2k + 1)\pi$ (hanging or rest position), $k \in \mathbb{Z}$. To avoid dealing with an infinite number of equilibria it will be assumed that x_1 belongs to S^1, and thus only two equilibrium points are distinguishable $x_{00} := (0, 0, 0)$, and $x_{\pi 0} := (\pi, 0, 0)$, corresponding to the pendulum at the inverted and the rest position, respectively, with zero disk velocity.

As a standard linear analysis reveals, system (1) is locally controllable at x_{00} such that a linear feedback of the form $u_l(x) = k_1x_1 + k_2x_2 + k_3x_3$, with $k_i \in \mathbb{R}$, locally stabilizes x_{00}. This linear feedback can be modified to preserve the symmetry of the system while maintaining the local properties of $u_l(x)$ at x_{00} as $u_s(x) = k_1 \sin x_1 + k_2x_2 + k_3x_3$. To limit the maximum amplitude of the control action to a certain value $u_{\max}$, the control law $u_s(x)$ is subject to a smooth saturation function, leading to the bounded continuous state feedback

$$u = u_{\max} \tanh(k_1 \sin x_1 + k_2x_2 + k_3x_3). \tag{2}$$

This control law (also considered in [13]) vanishes at x_{00} and $x_{\pi 0}$ preserving the equilibrium points of the open-loop system. However, additional operating points do arise, as will be shown next.

The closed-loop system actually has eight parameters: four related to the physical model and four associated to the controller. In the analysis performed in this paper the parameters of the model p_1, p_2, p_3 and ρ, and the value of the maximum control $u_{\max}$ are considered fixed, and the dynamical behavior of the closed-loop system is studied in terms of the control gains k_1, k_2 and k_3.

The controlled system (1)–(2) has odd $\mathbb{Z}_2$-symmetry, i.e. it is invariant under point reflection at the origin $(x_1, x_2, x_3) \to (-x_1, -x_2, -x_3)$. Moreover, it presents the same symmetry under rotations through π in x_1. As a consequence x_{00} and $x_{\pi 0}$ are always equilibrium points and some special kind of bifurcations are expected to occur (see [20]).

3.1. Bifurcation Analysis of Equilibria x_{00} and $x_{\pi 0}$

The local behavior of the closed-loop system is revealed by the characteristic polynomials of the linearized system at x_{00} and $x_{\pi 0}$

$$P_0(s) = s^3 + a_2s^2 + a_1s + a_0, \qquad P_\pi(s) = s^3 + a_2s^2 - a_1s - a_0,$$

respectively, where

$$\begin{aligned} a_0 &= \rho p_1 p_3 u_{\max}(k_3 - k_3^*), \\ a_1 &= p_3 u_{\max}(k_1 - k_1^*), \\ a_2 &= p_3 u_{\max}[k_2 - (k_3 - k_3^*)(1 + \rho)], \end{aligned}$$

with

$$k_1^* := \frac{p_1}{p_3 u_{\max}}, \qquad k_3^* := \frac{p_2}{p_3 u_{\max}}.$$

All three coefficients of the characteristic polynomials can be modified by varying the control gains k_1, k_2 and k_3, and thus five different bifurcation conditions can be obtained: a simple zero, a pair of purely imaginary zeros (Hopf bifurcation), a double zero, a pair of purely imaginary zeros plus a simple zero, and a triple zero. In the following, conditions on the control parameters leading to these bifurcations are derived.

3.1.1. *Simple Zero*

In the system under study a simple zero occurs when $a_0 = 0$ with $a_1 \neq 0$ and $a_2 \neq 0$, i.e. when $k_3 = k_3^*$ ($k_1 \neq k_1^*$ and $k_2 \neq 0$). This condition leads to a pitchfork bifurcation (referred to as P in the following) of equilibria x_{00} and $x_{\pi 0}$, and thus for $k_3 > k_3^*$ two antisymmetric equilibrium points associated to both x_{00} and $x_{\pi 0}$ do arise. The additional equilibria[2] are

$$x_{0+} := (x_{10}, 0, \hat{x}_{3e}), \qquad x_{0-} := (x_{10}, 0, -\hat{x}_{3e}),$$

and

$$x_{\pi+} := (x_{1\pi}, 0, \hat{x}_{3e}), \qquad x_{\pi-} := (x_{1\pi}, 0, -\hat{x}_{3e}),$$

respectively, where $\hat{x}_{3e}$ is the positive solution of

$$k_3^* x_{3e} - \tanh(k_3 x_{3e}) = 0. \tag{3}$$

The classification of the singularity is evident due to the symmetry of the bifurcation equation (3) and no reduction to the normal form is needed. It is also worth mentioning that this kind of bifurcation frequently arises in symmetrical systems with saturation [21].

3.1.2. *Hopf*

The equilibrium x_{00} (respectively, $x_{\pi 0}$) undergoes a Hopf bifurcation when $a_2 a_1 - a_0 = 0$, providing that $a_0 \neq 0$, $a_2 \neq 0$, and $a_1 > 0$ (respectively, $a_1 < 0$) . In terms of the control gains, this condition results in

$$k_1 = k_1^* \frac{k_2 - (k_3 - k_3^*)}{k_2 - (k_3 - k_3^*)(1 + \rho)}, \quad (k_3 \neq k_3^*)$$

valid for $k_1 > k_1^*$ (respectively, $k_1 < k_1^*$ when the equilibrium considered is $x_{\pi 0}$).

In the following, the Hopf bifurcation of x_{00} will be referred to as H1 and that of $x_{\pi 0}$ as H3. The stability of the emerging periodic solution may be determined by computing the curvature coefficient (see [13] for a calculation using a frequency-domain approach). Depending on the value of the control gains this bifurcation may present degeneracies giving rise to curves of saddle-node bifurcations of limit cycles.

[2] Notice that although the time derivative of the system vanishes at these points, the disk is spinning at constant speed either clockwise or counterclockwise.

3.1.3. *Double Zero*

A double-zero eigenvalue condition (referred to as DZ) is obtained when $a_0 = a_1 = 0$ and $a_2 \neq 0$, i.e. when the control gains take the values

$$k_1 = k_1^*, \quad k_3 = k_3^* \quad (k_2 \neq 0).$$

At these gains, the third eigenvalue of the linearized system is $\lambda = -u_{\max} p_3 k_2$. The system exhibits a double degeneracy for $k_2 \neq 0$, and for $k_2 > 0$ the nonzero eigenvalue λ is contractive.

Due to the symmetry of the system the unfolding of this codimension-two singularity is described by the Khorozov–Takens normal form (also found in the analysis of the 1:2 resonance [20]), as it will be shown next by the numerical unfolding of the bifurcation. Roughly speaking, this singularity introduces the dynamical phenomena of the previous bifurcations: the multiplicity of equilibrium solutions and the appearance of an oscillatory branch via the Hopf bifurcation mechanism. However, the complete characterization of the behavior of the system requires two bifurcation parameters and includes the appearance of homoclinic and heteroclinic orbits [20, 22, 23].

3.1.4. *Hopf and Simple Zero*

The equilibrium point x_{00} exhibits a Hopf plus a simple zero codimension-two bifurcation when $a_0 = a_2 = 0$ providing that $a_1 > 0$ (respectively, $a_1 < 0$ for the equilibrium point $x_{\pi 0}$). This condition occurs when the control gains take the values

$$k_3 = k_3^*, \quad k_2 = 0,$$

providing that $k_1 > k_1^*$ (respectively, $k_1 < k_1^*$ when the equilibrium point is $x_{\pi 0}$).

In this case, the interactions between the multiplicity of the equilibrium solutions and the oscillatory branch can lead to quasi-periodic motion and even chaotic motion when parameters are perturbed from the singularity condition. These complex dynamics are not examined in this article since the main objective is the stabilization of the pendulum at the inverted position.

3.1.5. *Triple Zero*

A triple-zero eigenvalue condition occurs when $a_0 = a_1 = a_2 = 0$, or equivalently

$$k_1 = k_1^*, \quad k_2 = 0, \quad k_3 = k_3^*.$$

The general unfolding of the triple-zero bifurcation is still under development. The generic case has been addressed in [24] giving a characterization of the triple-zero bifurcation up to codimension-four. The codimension-three unfoldings of this triple-zero linear degeneracy has been performed in [25, 26]. The triple-zero singularity with odd $\mathbb{Z}_2$-symmetry is analyzed in [27] on the Chua's circuit, while a partial unfolding of this bifurcation is given in [18] for the delayed control of the inverted pendulum on a cart. The triple-zero bifurcation contains all the previous singularities as particular cases. As in the case of the Hopf plus zero bifurcation, the complex behavior arising in this bifurcation is not addressed in this paper.

3.2. Bifurcation Analysis of Equilibria $x_{0\pm}$ and $x_{\pi\pm}$

As stated before, the additional equilibrium points $x_{0\pm}$ and $x_{\pi\pm}$ exist for $k_3 > k_3^*$ arising at the pitchfork bifurcation of x_{00} and $x_{\pi 0}$, respectively. The characteristic polynomials of the linearized system at $x_{0\pm}$ and $x_{\pi\pm}$ are

$$P_{0\pm}(s) = s^3 + \bar{a}_2 s^2 + \bar{a}_1 s + \bar{a}_0, \qquad P_{\pi\pm}(s) = s^3 + \bar{a}_2 s^2 - \bar{a}_1 s - \bar{a}_0,$$

respectively, where

$$\begin{aligned}
\bar{a}_0 &= \rho p_1 p_3 u_{\max} \alpha (k_3 - k_3^*/\alpha), \\
\bar{a}_1 &= p_3 u_{\max} \alpha (k_1 - k_1^*/\alpha), \\
\bar{a}_2 &= p_3 u_{\max} \alpha (k_2 - (k_3 - k_3^*/\alpha)(1+\rho)),
\end{aligned}$$

with $\alpha := \operatorname{sech}^2(k_3 \hat{x}_{3e})$ where $\hat{x}_{3e}$ is the positive solution of (3).

It can be proved that $\bar{a}_0 < 0$ for any $k_3 > k_3^*$. Therefore these equilibria may exhibit Hopf bifurcations only when $\bar{a}_2\bar{a}_1 - \bar{a}_0 = 0$ provided that $\bar{a}_1 > 0$ in the case of $x_{0\pm}$, or $\bar{a}_1 < 0$ in the case of $x_{\pi\pm}$. In terms of control gains, $\bar{a}_2\bar{a}_1 - \bar{a}_0 = 0$ is equivalent to set k_1 as

$$k_1 = \frac{k_1^*}{\alpha} \frac{k_2 - (k_3 - k_3^*/\alpha)}{k_2 - (k_3 - k_3^*/\alpha)(1+\rho)},$$

and provided that $k_2 > 0$, only $x_{\pi\pm}$ can exhibit Hopf bifurcations. In the following this bifurcation will be referred to as H2.

4. Analysis of the Dynamical Behavior

The main objective of the analysis of the system dynamics is to find appropriate values of the control gains k_1, k_2 and k_3 such that the inverted position x_{00} is stable with a large basin of attraction. By applying standard stability tests it is possible to find that local asymptotic stability of x_{00} is achieved when the control gains do satisfy

$$\begin{aligned}
&k_3 > k_3^*, \\
&k_2 > (1+\rho)(k_3 - k_3^*), \\
&k_1 > k_1^* \frac{k_2 - (k_3 - k_3^*)}{k_2 - (1+\rho)(k_3 - k_3^*)}.
\end{aligned} \tag{4}$$

Conditions (4) imply that k_1, k_2 and k_3 must be positive gains.

The singularity of higher complexity analyzed in this paper is the double-zero eigenvalue bifurcation, occurring simultaneously at x_{00} and $x_{\pi 0}$ when $k_1 = k_1^*$ and $k_3 = k_3^*$, provided that $k_2 \neq 0$. Therefore, the analysis of the dynamical behavior is carried out on a slice of the control parameters space with k_2 fixed at a positive value. For the numerical study $k_2 = 50$, and the following parameter values are used: $p_1 = 30\ \mathrm{s}^{-2}$, $p_2 = 0.0245\ \mathrm{s}^{-1}$, $p_3 = 0.0393\ \mathrm{s}^{-2}$, $\rho = 250$; the bound on the control amplitude was fixed at $u_{\max} = 60$. Notice that the value chosen for $u_{\max}$ sets a hard constraint, since the maximum

Table 1. Stability of the equilibrium points in the parameter plane $k_1 - k_3$ ($k_2 = 50$). See zones I–V in Figure 2. S: stable; U: unstable; $\times$: the equilibrium point does not exist.

Zone	x_{00}	x_{0+}	x_{0-}	$x_{\pi 0}$	$x_{\pi +}$	$x_{\pi -}$
I	S	U	U	U	U	U
II	S	U	U	U	S	S
III	U	U	U	U	S	S
IV	U	$\times$	$\times$	S	$\times$	$\times$
V	U	$\times$	$\times$	U	$\times$	$\times$

gravity effect represented by $p_1 = 30\ \mathrm{s}^{-2}$ is about 12 times greater than the maximum available input contribution $p_3 u_{\max} = 2.358\ \mathrm{s}^{-2}$.

4.1. Local Behavior: Stability Analysis

Let us analyze the stability of the system equilibria on the parameter plane $k_1 - k_3$. In accordance to the analysis performed in the previous section five different scenarios can be detected. These scenarios are delimited by pitchfork (P) and Hopf (H1, H2 and H3) bifurcation curves, as shown in Figure 2. In zones I, II and III there exist six physically distinguishable equilibrium points, and in zones IV and V only the trivial equilibria do exist. The stability of the equilibria is given in Table 1, revealing that the inverted position x_{00} is the unique stable equilibrium point in zone I. Therefore, the control objective can be achieved for a set of control gains in zone I if that set does not give rise to other attracting orbits (e.g. stable limit cycles). However, to fully characterize the basin of attraction of x_{00}, the global dynamical

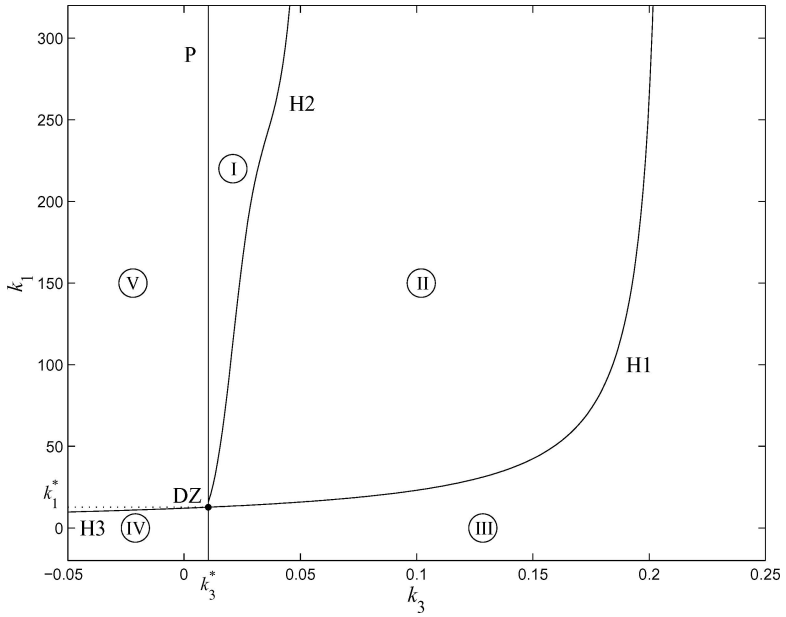


Figure 2. Zones in the parameter plane $k_1 - k_3$ exhibiting different stability of the equilibrium points (for $k_2 = 50$). H1, H2 and H3 are Hopf bifurcation curves, P is a pitchfork (simple-zero) bifurcation curve and the point DZ satisfies the condition for a double-zero bifurcation.

behavior of the system must be studied, focusing on the classification of bifurcations which can not be detected with the local stability analysis of the equilibria.

4.2. Global Behavior: Numerical Study

In a small neighborhood of x_{00} or $x_{\pi 0}$ the system behavior can be analyzed by reducing the dynamics to the center manifold associated to the double zero singularity [20, 22]. Nevertheless, to extend the analysis to a larger domain the whole nonlinear system should be considered. For this reason, a numerical analysis is performed using the continuation package AUTO [28] included in XPPAUT [29].

As it will be shown, the double-zero bifurcation acts as an organizing centre for the dynamics when the parameter plane $k_1 - k_3$ with $k_2 \neq 0$ is considered. For $k_2 = 50$, the equilibrium points x_{00} and $x_{\pi 0}$ undergo the double-zero bifurcation for $k_1 = k_1^* = 12.7226$ and $k_3 = k_3^* = 0.01039$. Figure 3 depicts the numerical unfolding of this codimension-two bifurcation experienced simultaneously by x_{00} and $x_{\pi 0}$, and also other global nonlinear phenomena associated to interactions between these double-zero bifurcations. This bifurcation diagram and the phenomena summarized in Table 2 are described in the following sections.

4.2.1. *Codimension-One Bifurcation Points*

For $k_3 < k_3^*$ only two equilibrium points do exist: the rest position $x_{\pi 0}$ and the inverted position x_{00}. For $k_3 > k_3^*$ there exist six equilibria due to the pitchfork bifurcation (P) of x_{00} and $x_{\pi 0}$, given rise to x_{0+}, x_{0-} , and $x_{\pi +}$, $x_{\pi -}$, respectively. The equilibrium points for $k_3 > k_3^*$ on the plane $x_1 - x_3$ are depicted in Figure 4, where the stability of the equilibrium points and the direction of the flow lines may change depending on the parameter values. Notice that the segments A–A′ and B–B′ are homologous due to the periodicity of variable x_1 ($x_1 \in S^1$).

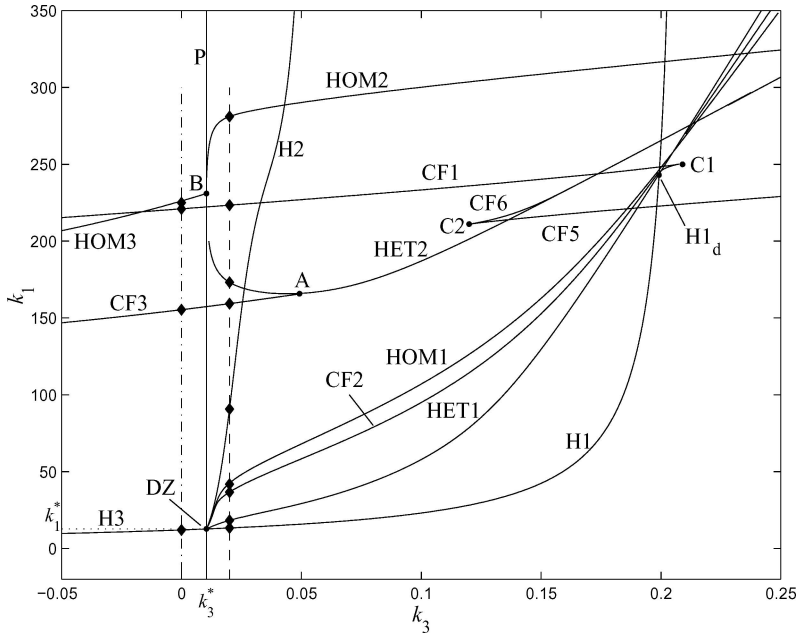


Figure 3. Numerical unfolding of the double-zero bifurcations (DZ) exhibited simultaneously by x_{00} and $x_{\pi 0}$ when $k_1 = k_1^* = 12.7226$ and $k_3 = k_3^* = 0.01039$ for $k_2 = 50$ (see Table 2 for a brief description of the labels).

Table 2. Symbols used to represent the different bifurcations of the system.

Symbol	Phenomenon
P	Pitchfork bifurcation of x_{00} and $x_{\pi 0}$ (simultaneous)
H1	Hopf bifurcation of x_{00}
H2	Hopf bifurcation of $x_{\pi+}$ and $x_{\pi-}$ (simultaneous)
H3	Hopf bifurcation of $x_{\pi 0}$
HOM1	Homoclinic bifurcation to $x_{\pi 0}$ (eight figure)
HOM2	Homoclinic bifurcation to x_{0+} and x_{0-} (simultaneous)
HOM3	Homoclinic bifurcation to x_{00}
HET1	Heteroclinic bifurcation involving x_{0+} and x_{0-} (small cycle)
HET2	Heteroclinic bifurcation involving x_{0+} and x_{0-} (big cycle)
$H1_d$	Degenerate Hopf bifurcation of x_{00}
CF1 to CF6	Cyclic fold bifurcations
C1 and C2	Cusps of cyclic fold bifurcations
A	Coalescence of CF3 at the heteroclinic HET2
B	Interaction between homoclinic (HOM3) and pitchfork (P) bifurcations
DZ	Double-zero bifurcation of equilibria x_{00} and $x_{\pi 0}$ (simultaneous)

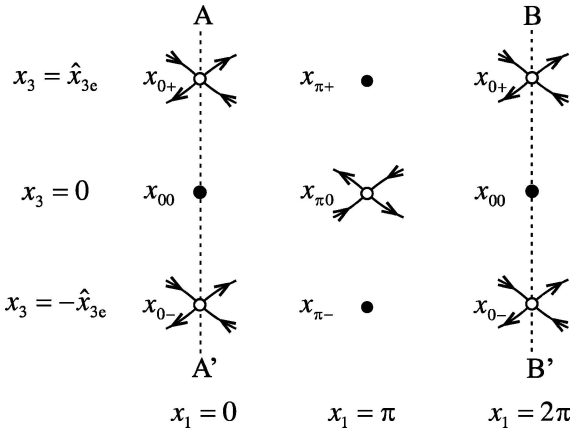


Figure 4. Equilibrium points on the plane $x_1 - x_3$ for $k_3 > k_3^*$.

In order to describe the bifurcation diagram, let us consider first $k_3 = 0.02$ (see the dashed line in Figure 3). The different behaviors obtained when k_1 is varied are shown in Figure 5 where the period of the limit cycles is plotted against parameter k_1. Beginning from the left, the first detected phenomenon is a Hopf bifurcation (H1) experienced by x_{00} when $k_1 = 13.365$ (see also the curve H1 in Figure 3). The equilibrium point x_{00} changes from unstable to stable as k_1 passes through this critical value. In addition, an unstable limit cycle (γ_0^u) appears and its amplitude grows until it collides with x_{0+} and x_{0-} for $k_1 = 18.303$ leading to a heteroclinic bifurcation (or homoclinic cycle). At this point (HET1) the unstable manifold of x_{0+} is connected to the stable manifold of x_{0-}. Notice that the period of this cycle grows asymptotically to infinity. The cycle in the proximity of the bifurcation point ($k_1 = 18.30$) is shown in Figure 6.

A second local phenomenon is a Hopf bifurcation (H2) exhibited by $x_{\pi+}$ and $x_{\pi-}$ at $k_1 = 90.699$. For $k_1 < 90.699$, $x_{\pi+}$ and $x_{\pi-}$ are stable and they are surrounded by antisymmetric unstable limit cycles

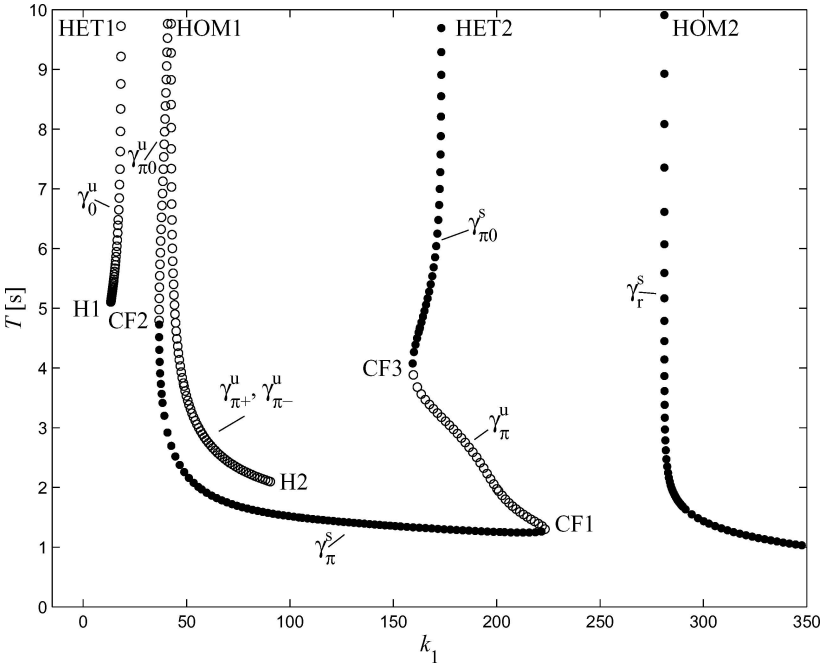


Figure 5. Continuation curves for $k_3 = 0.02$ ($k_2 = 50$). (•) Stable limit cycles; (○) unstable limit cycles.

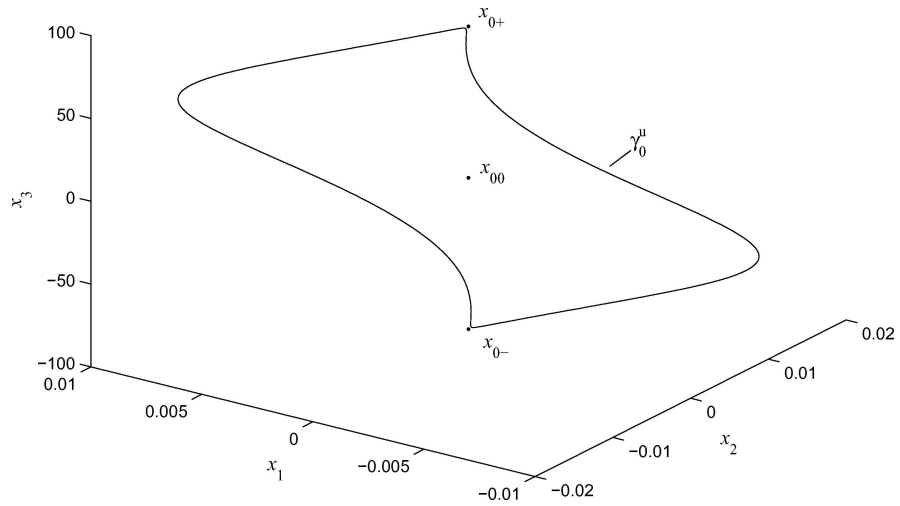


Figure 6. Cycle γ_0^u near the heteroclinic bifurcation HET1 ($k_1 = 18.30$, $k_2 = 50$ and $k_3 = 0.02$).

$\gamma_{\pi+}^u$ and $\gamma_{\pi-}^u$ (see Figure 7a). These cycles do collide with the equilibrium point $x_{\pi 0}$ at $k_1 = 42.488$ when a double saddle connection or homoclinic bifurcation occurs (HOM1). Both cycles, near the typical eight figure, are shown in Figure 7b. For $k_1 < 42.488$ they form a unique unstable cycle (see Figure 7c) which coalesces with a stable one (γ_π^s) at the cyclic fold bifurcation CF2 for $k_1 = 36.700$ (see Figure 7d).

For $k_1 = 159.331$ a cyclic fold bifurcation (CF3) occurs, and thus stable and unstable cycles ($\gamma_{\pi 0}^s$ and γ_π^u, respectively) do arise for $k_1 > 159.331$. The stable one ($\gamma_{\pi 0}^s$) coalesces in an heteroclinic bifurcation (HET2) when $k_1 = 173.228$, while the unstable one (γ_π^u) collapses in a cycle fold (CF1)

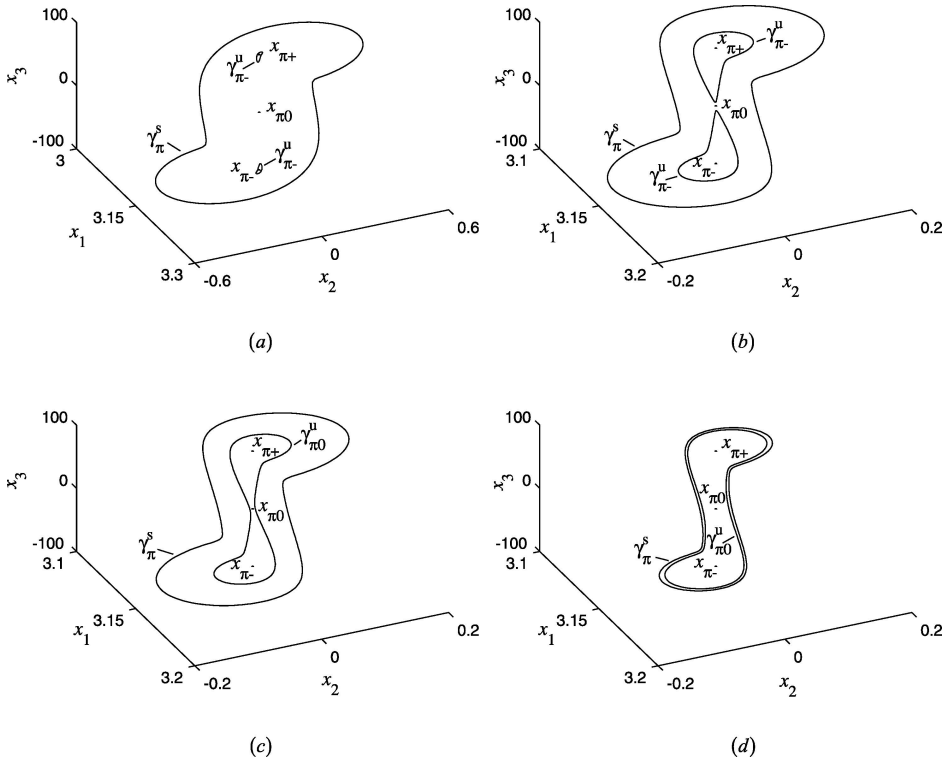


Figure 7. Evolution of the limit cycles $\gamma^u_{\pi+}$ and $\gamma^u_{\pi-}$. Limit cycles obtained with $k_2 = 50$, $k_3 = 0.02$ and: (a) $k_1 = 80.9926$; (b) $k_1 = 43.0132$; (c) $k_1 = 41.1747$; (d) $k_1 = 36.7497$.

with γ^s_π at $k_1 = 223.386$. Observe that three periodic solutions ($\gamma^s_{\pi 0}$, γ^u_π and γ^s_π) coexist for k_1 in the interval comprised by CF3 and HET2 (see Figure 8).

Finally, a homoclinic bifurcation (HOM2) occurs at $k_1 = 281.069$. Due to this global phenomenon, a stable cycle (γ^s_r) arises for increasing values of k_1, corresponding to rotations of the pendulum around the pivot point.

Continuing with the unfolding of Figure 3, let us consider next $k_3 = 0$ (see the dash-dot line in that figure). Since $k_3 < k_3^*$ the only equilibrium points are x_{00} and $x_{\pi 0}$. The bifurcation diagram, obtained when varying k_1, is shown in Figure 9. The equilibrium point $x_{\pi 0}$ exhibits a Hopf bifurcation (H3) for $k_1 = 12.0945$ and a stable limit cycle (γ^s_π) appears for increasing values of k_1. For $k_1 = 155.385$ a cyclic fold bifurcation (CF3) occurs and a stable and unstable cycles ($\gamma^s_{\pi 0}$ and γ^u_π, respectively) do appear; therefore three cycles coexist for higher values of k_1 as depicted in Figure 10. The stable cycle ($\gamma^s_{\pi 0}$) undergoes a homoclinic bifurcation (HOM3) at $k_1 = 225$, since it reaches the equilibrium point x_{00}, and it is destroyed by this bifurcation. Finally, the remaining unstable and stable cycles (γ^u_π and γ^s_π, respectively) coalesce in a cyclic fold bifurcation (CF1) at $k_1 = 221.032$.

By numerical continuation of the codimension-one bifurcation points described above, most of the bifurcation curves in Figure 3 are obtained. An additional local bifurcation phenomena appearing in that picture is a degenerate Hopf bifurcation at curve H1 when $k_1 = 243.052$ and $k_3 = 0.1992$. This singularity leads to a first stability coefficient equal to zero, and the analytical determination of the type of bifurcation and its unfolding, requires a fifth-order expansion. At this point ($H1_d$ in Figure 3) a cyclic fold bifurcation CF4 arises, which collapses at HET1 for larger values of k_1 and k_3 in the vicinity of $H1_d$, as shown in Figure 11.

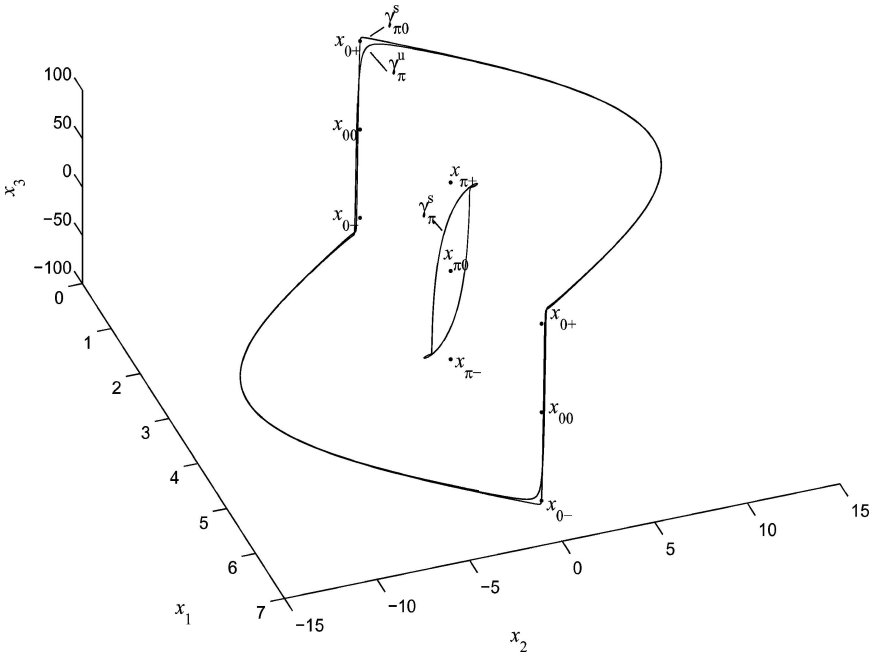


Figure 8. Limit cycles obtained with $k_1 = 170.7$, $k_2 = 50$ and $k_3 = 0.02$.

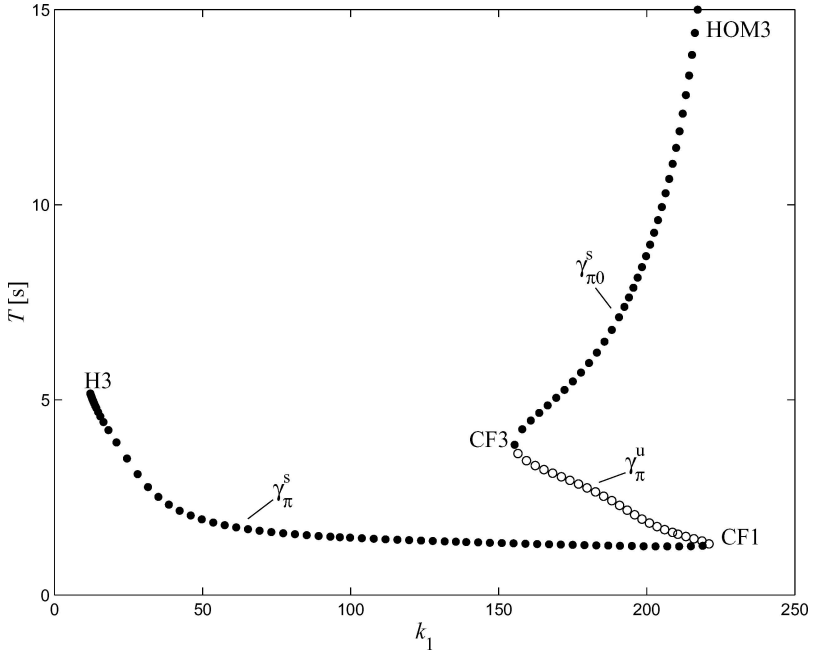


Figure 9. Continuation curves for $k_3 = 0$ ($k_2 = 50$). (•) Stable limit cycles; (○) unstable limit cycles.

The bifurcation diagram of Figure 3 allow us to classify the double-zero singularity as well as to describe other global phenomena. It is worth mentioning that the continuation of heteroclinic bifurcations HET1 and HET2 is very difficult to perform due to numerical issues and they have been approximated by continuing a cycle of a large period.

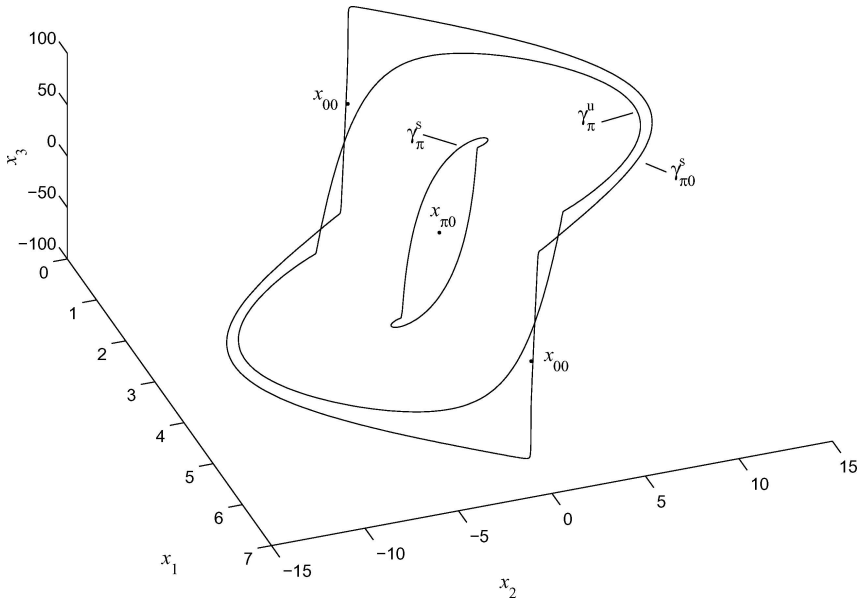


Figure 10. Cycles obtained with $k_1 = 170.76$, $k_2 = 50$ and $k_3 = 0$.

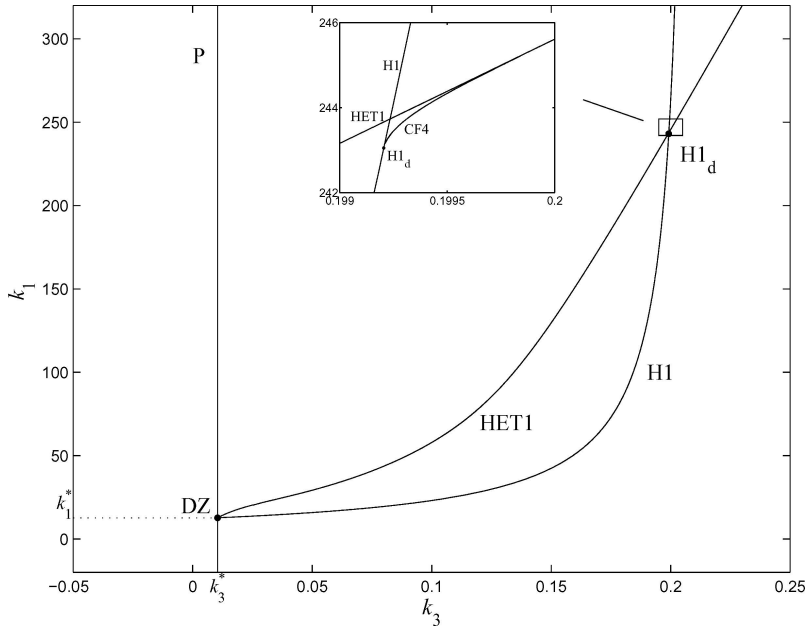


Figure 11. Degenerate Hopf bifurcation ($H1_d$) at curve H1 for $k_1 = 243.052$ and $k_3 = 0.1992$ with $k_2 = 50$ (see Table 2 for a brief description of the labels).

4.2.2. *Classification of the Double-Zero Singularity*

Bifurcation curves P, H1, H2, H3, HET1, HOM1, and CF2 meet at the double-zero singularity point $k_1 = k_1^*$ and $k_3 = k_3^*$. The classification of this singularity is derived from the analysis of the unfolding in the neighborhood of the double-zero point. Bifurcation curves H2, H3, HOM1 and CF2 are associated

to $x_{\pi 0}$ while curves H1 and HET1 are connected to codimension-one bifurcations of x_{00}. Notice that x_{00} and $x_{\pi 0}$ exhibit pitchfork bifurcation (curve P) simultaneously. Figure 12 details the bifurcation curves associated to the double-zero bifurcation experienced by x_{00}, whereas Figure 13 depicts those corresponding to $x_{\pi 0}$.

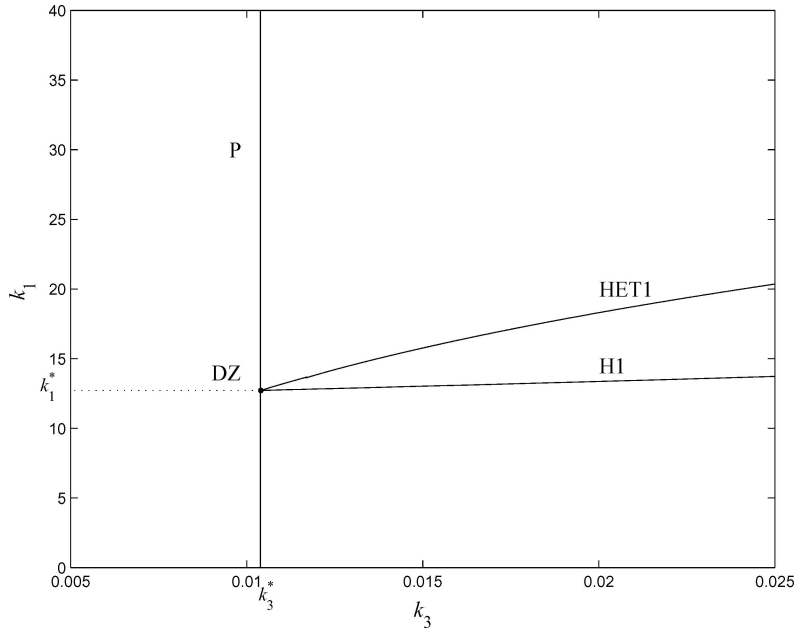


Figure 12. Numerical unfolding of the double-zero bifurcation (DZ) exhibited by x_{00} at $k_1 = k_1^* = 12.7226$, $k_3 = k_3^* = 0.01039$ and $k_2 = 50$ (case $s = 1$ in the normal form; see Table 2 for a brief description of the labels).

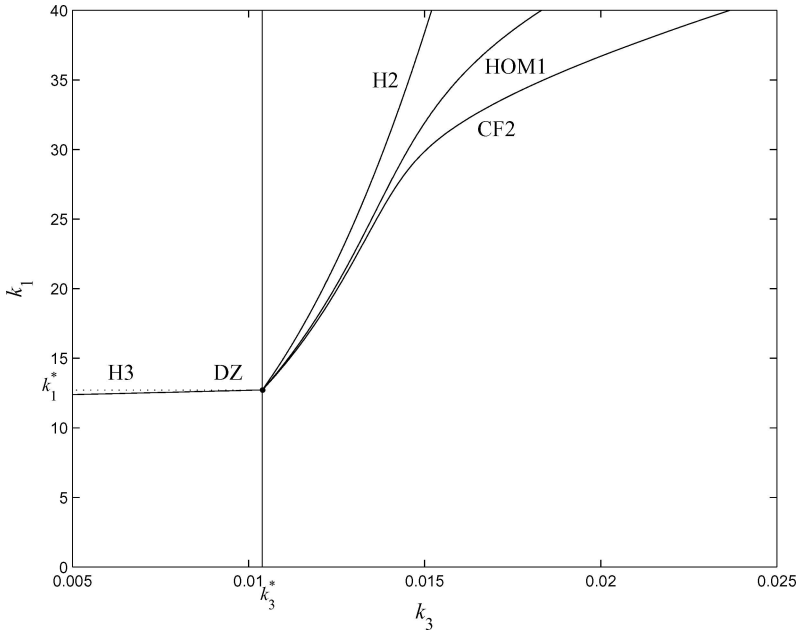


Figure 13. Numerical unfolding of the double-zero bifurcation (DZ) exhibited by $x_{\pi 0}$ at $k_1 = k_1^* = 12.7226$, $k_3 = k_3^* = 0.01039$ and $k_2 = 50$ (case $s = -1$ in the normal form; see Table 2 for a brief description of the labels).

The Khorozov–Takens normal form is given by

$$\dot{x} = y,$$
$$\dot{y} = \mu_1 x + \mu_2 y + s x^3 - x^2 y,$$

with $s = 1$ or $s = -1$. The unfolding of this double-zero bifurcation exhibited by the origin when $\mu_1 = \mu_2 = 0$ can be found, for example, in [20] concerning the analysis of the resonance 1:2, in [22] regarding the study of the double-zero eigenvalue with cubic symmetry, or in [23]. A direct comparison between the unfoldings of Figures 12 and 13 and those in the mentioned references reveals that the unfolding associated to x_{00} (Figure 12) corresponds to $s = 1$ in the Khorozov–Takens normal form, and that of $x_{\pi 0}$ (Figure 13) to $s = -1$.

4.2.3. *Other Global Phenomena*

As can be appreciated in Figure 3, the system exhibits other global phenomena which cannot be detected by the local analysis of the Khorozov–Takens bifurcation. The homoclinic cycle created at the bifurcation HET2 changes its stability when the cyclic fold bifurcation CF3 meets the curve HET2 (point A). The pitchfork P and the homoclinic HOM3 bifurcation curves interact giving rise to the homoclinic and heteroclinic bifurcation curves HOM2 and HET2 (point B).

Two cusps of cyclic fold bifurcations are detected at points C1 and C2. The point C1 is the cusp of the curves CF1 and CF2, while C2 is the cusp of CF5 and CF6. The cyclic fold bifurcation curve CF6 coalesces with the heteroclinic curve HET2.

Evidence of complex dynamical behavior in the neighborhood of the curves HET2 and HOM1 when $k_1 > 250$ and $k_3 > 0.2$ has been found. This behavior can be related to additional global phenomena associated to the triple-zero bifurcation and reflection symmetry (see [30] where complex balancing motions of an inverted pendulum are studied). This issue is not pursued further in this paper.

5. Basin of Attraction of the Inverted Position

An important consequence of the numerical characterization of the dynamical behaviour of the system is that for k_1 and k_3 values comprised within region (a) in Figure 14 the equilibrium point x_{00} is the unique attractor of the system and thus the asymptotic stabilization of the pendulum at the inverted position can be achieved. Extensive simulation results allow us to conjecture on the existence of an attractive two-dimensional manifold containing all the equilibrium points (some numerical simulations can be found in [31]). Based on this hypothesis, qualitative phase portraits giving a graphical interpretation of the distinctive dynamical behaviors of the system with parameter values in regions (a)–(d), are shown in Figure 15. The aim of this description is not to provide a theoretical justification of the stabilization procedure but only to illustrate the numerical results.

The qualitative phase portrait for the system with control parameters in region (a) is shown in Figure 15a. The only stable equilibrium point (black point) is x_{00} and no other attractors do appear. The unstable equilibrium points (white points) x_{0+} , x_{0-} and $x_{\pi 0}$ are saddles, and $x_{\pi +}$ and $x_{\pi -}$ are foci (saddle-foci in the three-dimensional system). Thus, in region (a) the system exhibits the 'best' global behavior, in the sense that the basin of attraction of x_{00} is larger than the basin of attraction in regions (b), (c) and (d). Moreover, this study suggests that the stabilization is almost global.

When k_1 and k_3 are varied in such a way that the curve H2 in Figure 14 is crossed from region (a) to region (b), the unstable equilibria $x_{\pi +}$ and $x_{\pi -}$ undergo a Hopf bifurcation, changing their stability. The

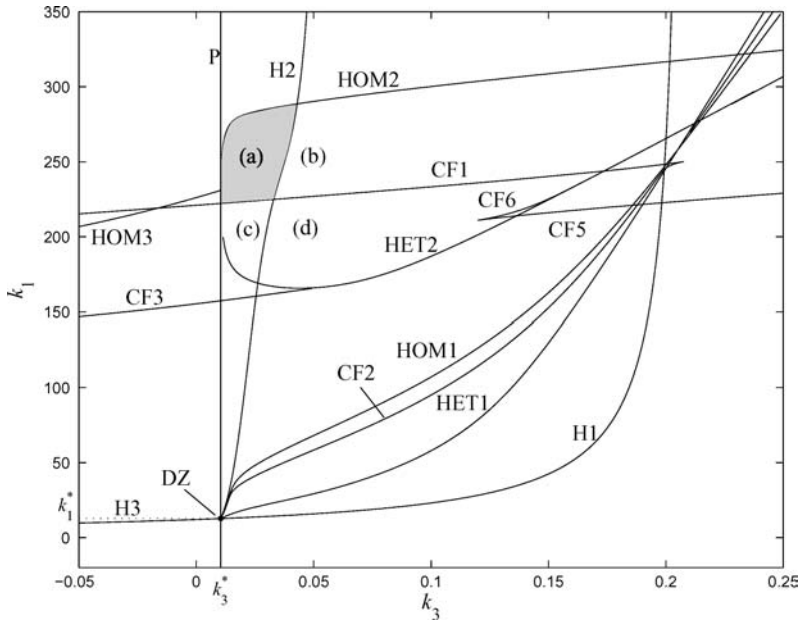


Figure 14. The equilibrium x_{00} (inverted position with zero disk velocity) is the unique attractor of the system when k_1 and k_3 belong to region (a) (see Table 2 for a brief description of the labels).

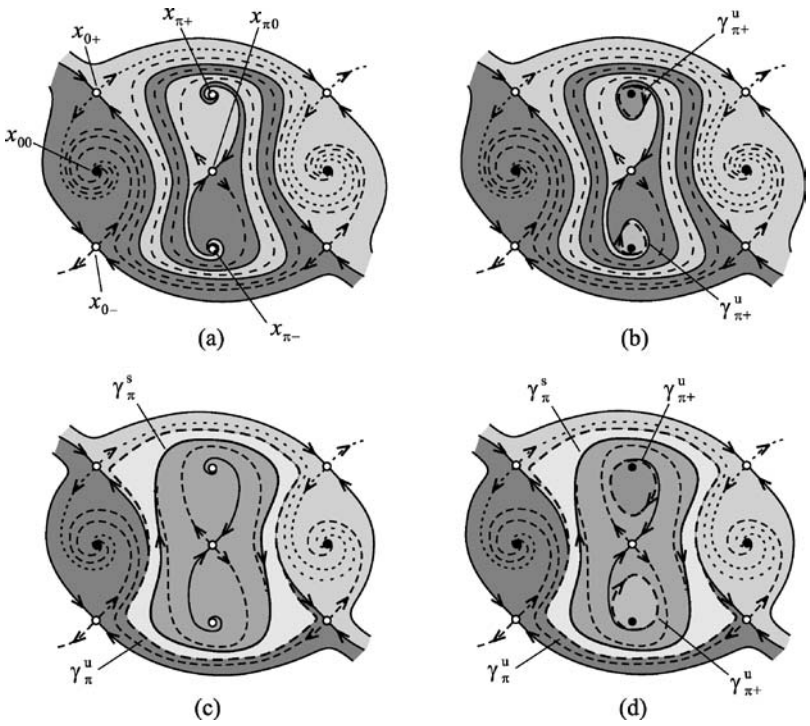


Figure 15. Phase portraits on a two-dimensional manifold for gains values in regions (a), (b), (c) and (d) in Figure 8.

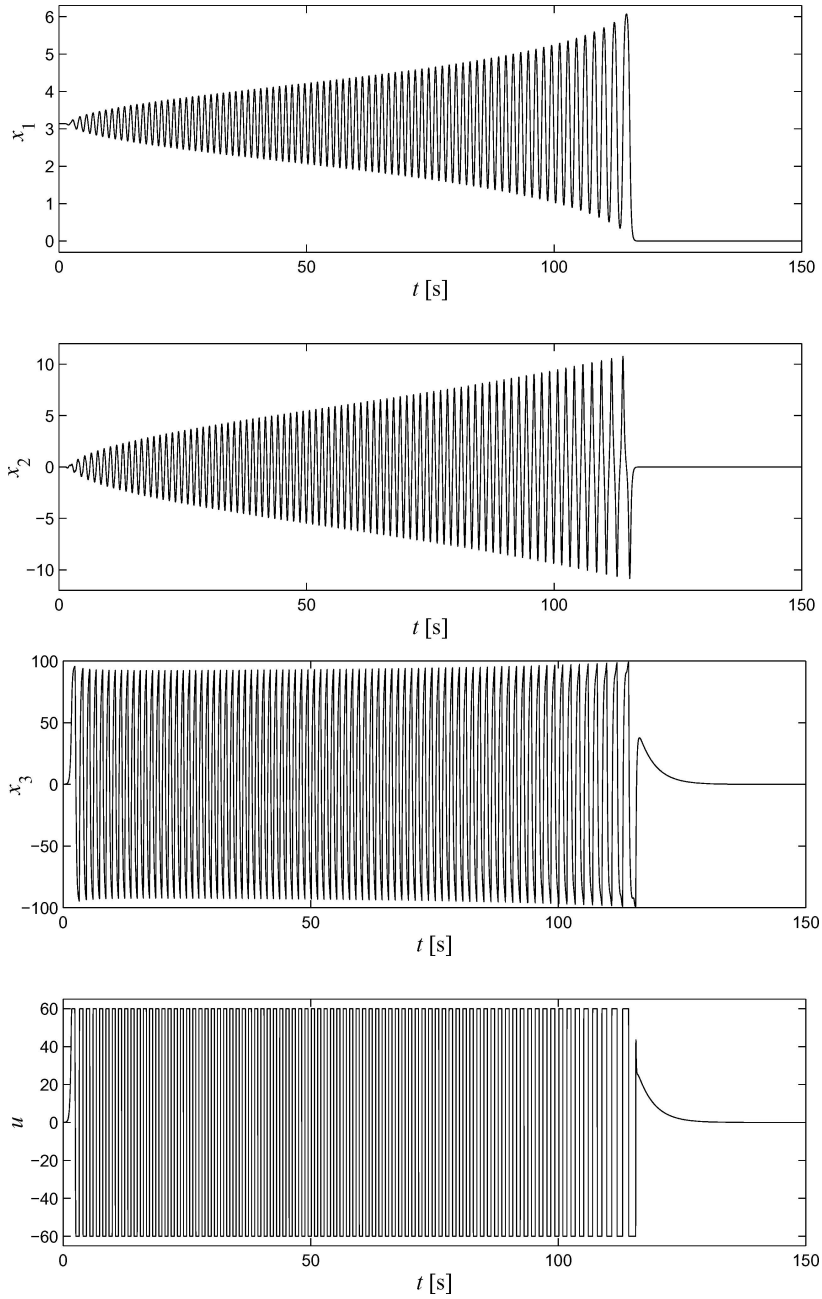


Figure 16. Evolution of the states of the system and control action for gains $k_1 = 260$, $k_2 = 50$ and $k_3 = 0.02$. The initial condition is $x = (\pi, 0, 0.1)$. The pendulum is stabilized at the inverted position with zero disk velocity at $t \approx 130$ s.

two emerging unstable limit cycles ($\gamma^u_{\pi+}$ and $\gamma^u_{\pi-}$) restrict the domain of attraction of x_{00}, as depicted in Figure 15b. In region (c) a pair of limit cycles (γ^u_π and γ^s_π) of opposed stability appear due to the cyclic fold bifurcation CF1, drastically reducing the basin of attraction of x_{00} (see Figure 15c) and isolating the rest position $x_{\pi 0}$. Therefore, it is impossible to reach the inverted position from the neighborhood of the hanging position, since the trajectories are captured by a stable limit cycle. When

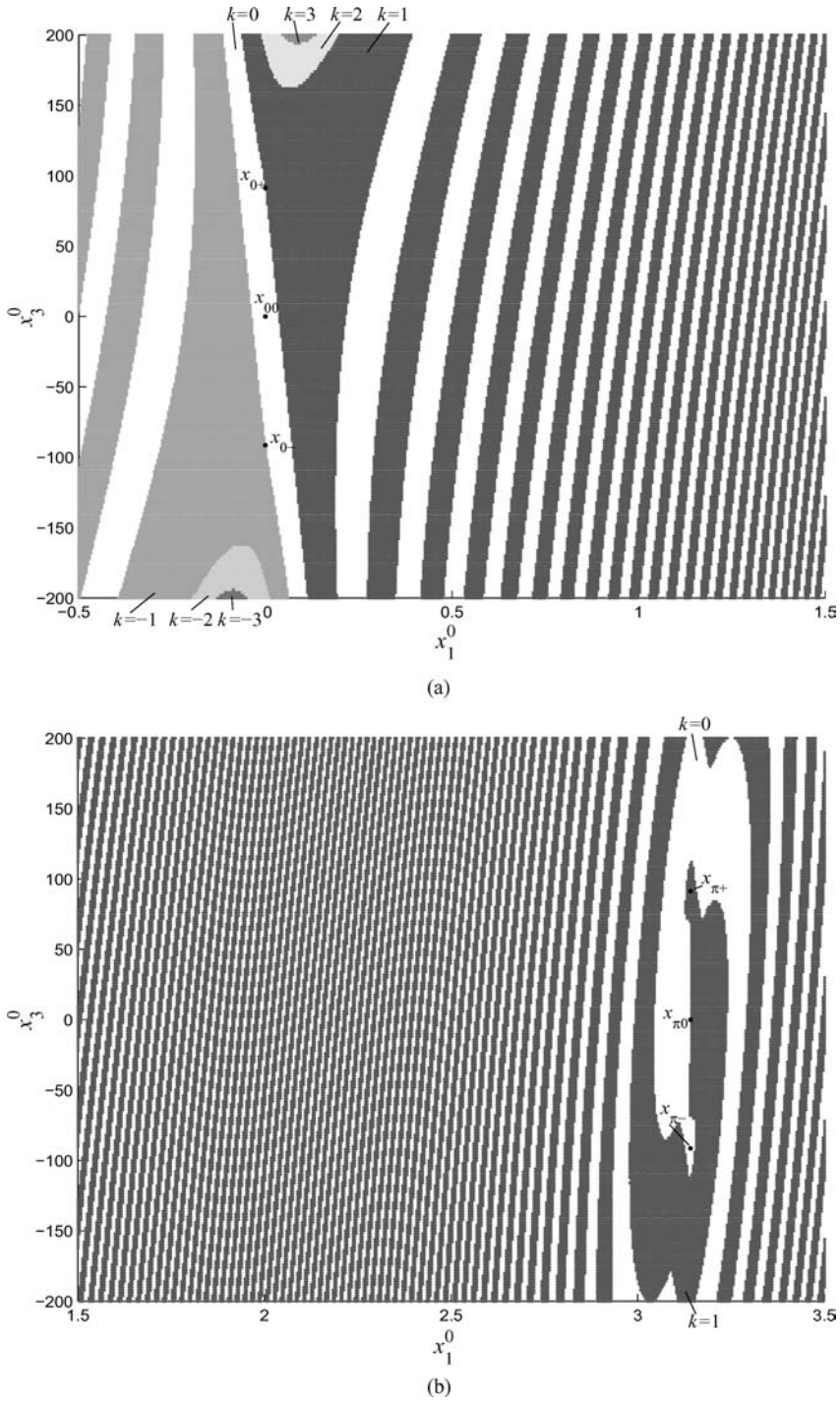


Figure 17. Basin of attraction of the inverted position on the plane $x_2 = 0$. Initial conditions (x_1^0, x_3^0) are shaded according to the value of $x_1 \in \mathbb{R}$ (not restricted to S^1) where the trajectory ends up. The final value is $x = (2k\pi, 0, 0)$ with $k \in \mathbb{Z}$. (a) $x_1^0 \in [-0.5, 1.5]$, (b) $x_1^0 \in [1.5, 3.5]$, and $x_3^0 \in [-200, 200]$ on both cases.

crossing the curve H2 in Figure 14 towards region (d), two additional small antisymmetric cycles ($\gamma^u_{\pi+}$ and $\gamma^u_{\pi-}$) surrounding $x_{\pi+}$ and $x_{\pi-}$ do appear (Figure 15d).

Figure 16 shows a numerical simulation of the controlled system for control gains lying inside region (a) of Figure 14 ($k_1 = 260$, $k_2 = 50$, $k_3 = 0.02$). The initial condition is $x = (\pi, 0, 0.1)$ (close to the rest position) and after a series of swings of increasing amplitude the pendulum is stabilized at the inverted position as desired.

The basin of attraction of the upright position was studied numerically for a set of equally spaced initial conditions $(x_1^0, 0, x_3^0)$ on the plane $x_2 = 0$, as shown in Figure 17 for $x_1^0 \in [-0.5, 3.5]$ and $x_3^0 \in [-200, 200]$, where the interval for x_1^0 was splitted into $x_1^0 \in [-0.5, 1.5]$ (Figure 17a) and $x_1^0 \in [1.5, 3.5]$ (Figure 17b). To provide more details on the dynamical behavior, x_1 is displayed as a real variable ($x_1 \in \mathbb{R}$). Thus, in that figure each initial condition is denoted with a shade according to the value of x_1 where the trajectory end up, i.e. $x_1 = 2k\pi$, $k \in \mathbb{Z}$. It is worth mentioning that all the trajectories finish at the inverted position with zero velocity of the disk (the equilibrium point x_{00} when considering $x_1 \in S^1$).

6. Conclusions

In this paper, the closed-loop dynamics of an underactuated mechanical system is analyzed using tools from bifurcation theory. The purpose of this study is to obtain feedback gains values providing the "best" global behavior in the sense that the upright position is stabilized with a large basin of attraction. After an analytical local bifurcation study, a numerical bifurcation analysis is performed to characterize the dynamical behavior over a larger domain. A partial bifurcation diagram for two control parameters has been obtained showing that the dynamics is organized in a double zero bifurcation with rotational symmetry, known as Khorozov–Takens bifurcation. This bifurcation diagram reveals that exists a region in the space of control gains where the inverted position has a large basin of attraction, and a detailed analysis allow us to presume that selecting gain values within that region an almost global stabilization of the inverted position is achieved.

This methodology can be employed to analyze the behavior of other controllers reported in the literature, taking into account saturation effects. However, the extension of this approach to larger dimensional systems may be difficult and laborious to apply.

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